Millimeter Wave MIMO Communication

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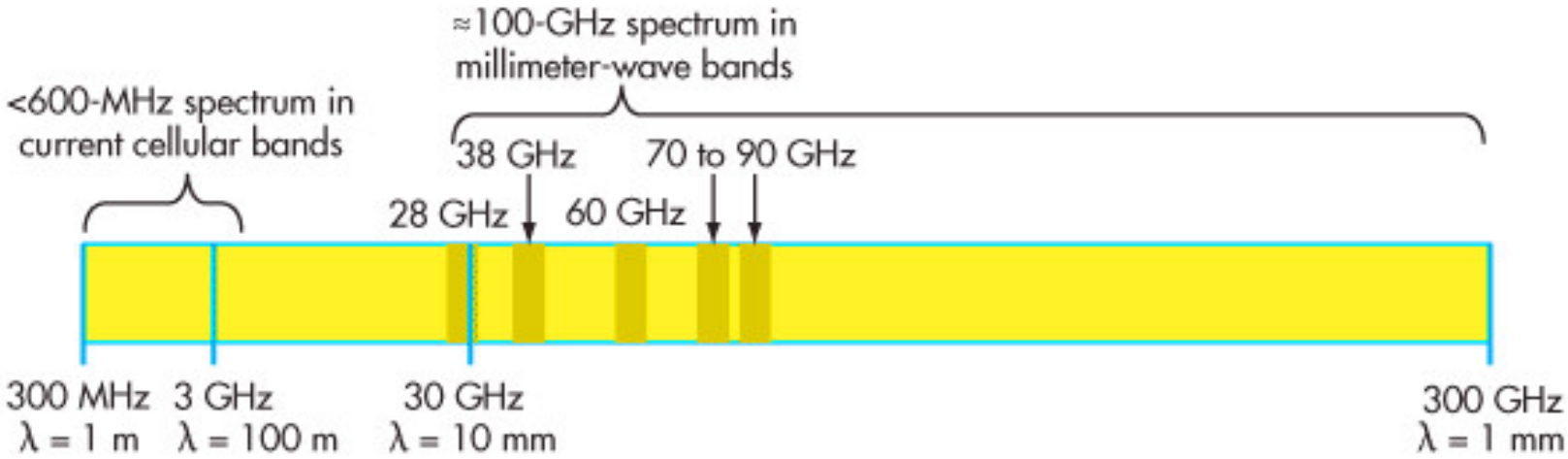
June 27, 2016

Overview

1. Introduction to mmWave
   - Why mmWave?
   - Channel Model
     - Large Scale Parameters
   - Architecture

2. Hybrid Precoding Design
   - State-of-the-Art

Why mmWave?



- More bandwidth availabe in mmWave bands
- High throughput in small geographic areas

Image source: http://electronicdesign.com/

mmWave Challenges

- Sensitive to blockages, such as outer brick walls of buildings, cause severe penetration loss
- Variable propagation laws
  - Weaker diffraction, hence LOS propagate as in free space
  - LOS path loss exponent close to 2, while the NLOS path losses are generally larger and more dependent on the scattering environment
- Sparsity in channel
  - Few multipath components
  - Small angular spread
- Power-limited system [not BW limited]

Channel characteristics

- Free space propagation path loss [Friis' law]

$$P_r = G_r G_t \left(\frac{\lambda}{4\pi d}\right)^2 P_t$$

- Path loss in inversly propertional to wavelength squared, implies non-directional mmWave will experience higher path loss
- However, the scaling gains of antennas more compensates than the path loss at mmWave frequencies

$$G_r, G_t \propto \lambda^{-2}$$

Channel Model

Channel impulse response:

$$h(t, \vec{\Theta}, \vec{\Phi}) = \sum_{l_1=1}^{L_{AOD}} \sum_{l_2=1}^{L_{AOA}} \sum_{n=1}^{N} \sum_{m=1}^{M_n} a_{m,n,l_1,l_2} e^{j\varphi_{m,n}} \cdot \delta(t - \tau_{m,n}) \cdot \delta(\vec{\Theta} - \vec{\Theta}_{l_1}) \cdot \delta(\vec{\Phi} - \vec{\Phi}_{l_2})$$

Path loss model:

$$P_r[dBm] = P_t + G_t + G_r - PL[dB]$$
$$PL_{NLOS}[dB] = 61.4 + 34 \log_{10}(d) + \chi_\sigma, \quad d \geq 1 \text{ m}$$

Time clusters and spatial AOD and AOA lobes

$$N \sim DU[1, 6]$$
$$L_{AOD} \sim \min\left\{L_{max}, \max\left\{1, \min\left\{A, N\right\}\right\}\right\}$$
$$L_{AOA} \sim \min\left\{L_{max}, \max\left\{1, \min\left\{B, N\right\}\right\}\right\}$$

and: $A \sim \text{Poisson}(\mu_{AOD} + 0.2)$

$B \sim \text{Poisson}(\mu_{AOA} + 0.1)$

Channel Model

Intra cluster subpath excess delays:

$$\rho_{m,n}(B_{bb}) = \left\{ \frac{1}{B_{bb}} \times (m-1) \right\}^{1+X}$$

Cluster excess delays:

$$\tau_n'' \sim \text{Exp}(\mu_\tau)$$
$$\Delta\tau_n = \text{sort}(\tau_n'') - \min(\tau_n'')$$
$$\tau_n = \begin{cases} 0, & n = 1 \\ \tau_{n-1} + \rho_{M_{n-1},n-1} + \Delta\tau_n + 25, & n = 2, ..., N \end{cases}$$

Cluster powers:

$$P_n' = \overline{P}_0 e^{-\frac{\tau_n}{\Gamma}} 10^{\frac{Z_n}{10}}$$
$$P_n = \frac{P_n'}{\sum_{k=1}^{k=N} P_k'} \times P_r[mW]$$
$$Z_n \sim N(0, 3 \text{ dB}), n = 1, 2, ...N$$

Channel Model

Subpath powers:

$$\Pi'_{m,n} = \overline{\Pi}_0 e^{-\frac{\rho_{m,n}}{\gamma}} 10^{\frac{U_{m,n}}{10}}$$
$$\Pi_{m,n} = \frac{\Pi'_{m,n}}{\sum_{k=1}^{k=N} \Pi'_{k,n}} \times P_n[mW]$$
$$U_{m,n} \sim N(0, 6 \text{ dB})$$
$$m = 1, 2, ..., M_n \quad , \quad n = 1, 2, ...N$$

Subpath phases:

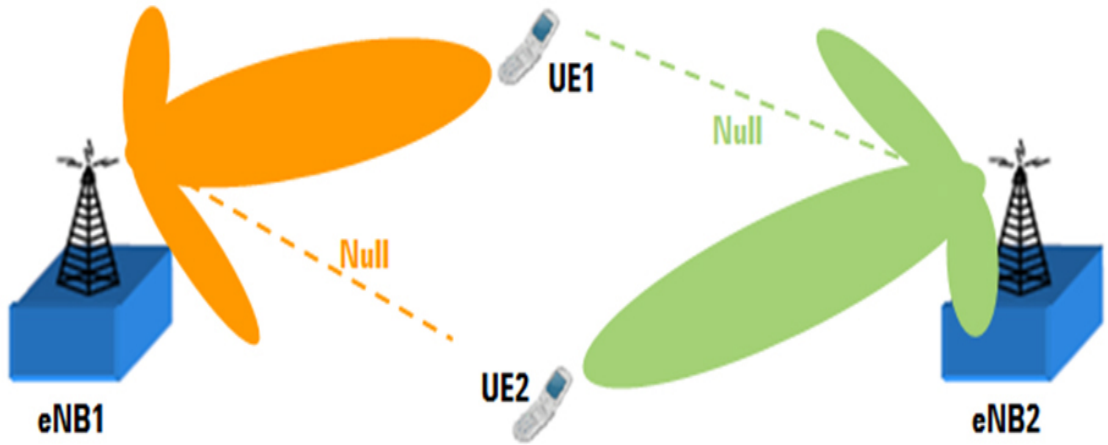
$$\varphi_{m,n} = \varphi_{1,n} + 2\pi f \rho_{m,n}$$
$$\varphi_{1,n} \sim U(0, 2\pi)$$
$$m = 2, ..., M_n, n = 1, 2, ..., N$$

Mean AOA and AOD in azimuthal domain:

$$\theta_i \sim DU[\theta_{min}, \theta_{max}] \quad , \quad i = 1, 2, ..., L$$
$$\theta_{min} = \frac{360(i-1)}{L} \quad , \quad \theta_{max} = \frac{360i}{L}$$

Why MIMO at mmWave?

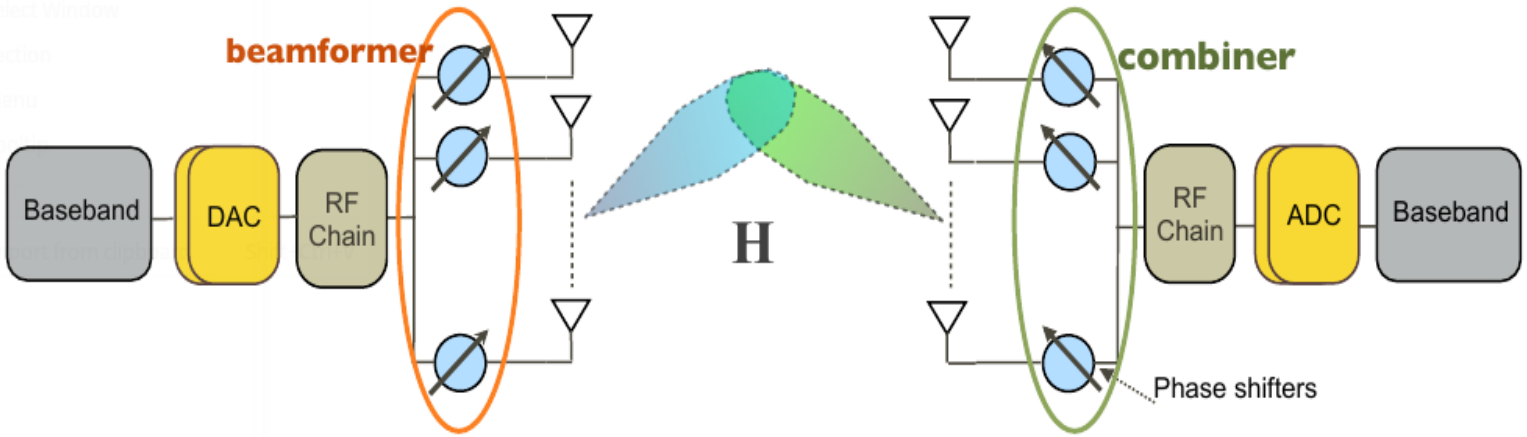
- Beamforming
- Spatial multiplexing & Beamforming
- Array gain
- Other MIMO



SM is for bandlimited systems; Beamforming is for power-limited systems

Image source:
http://www.wirelessdesignmag.com/sites/wirelessdesignmag.com/files/Agilent2.jpg

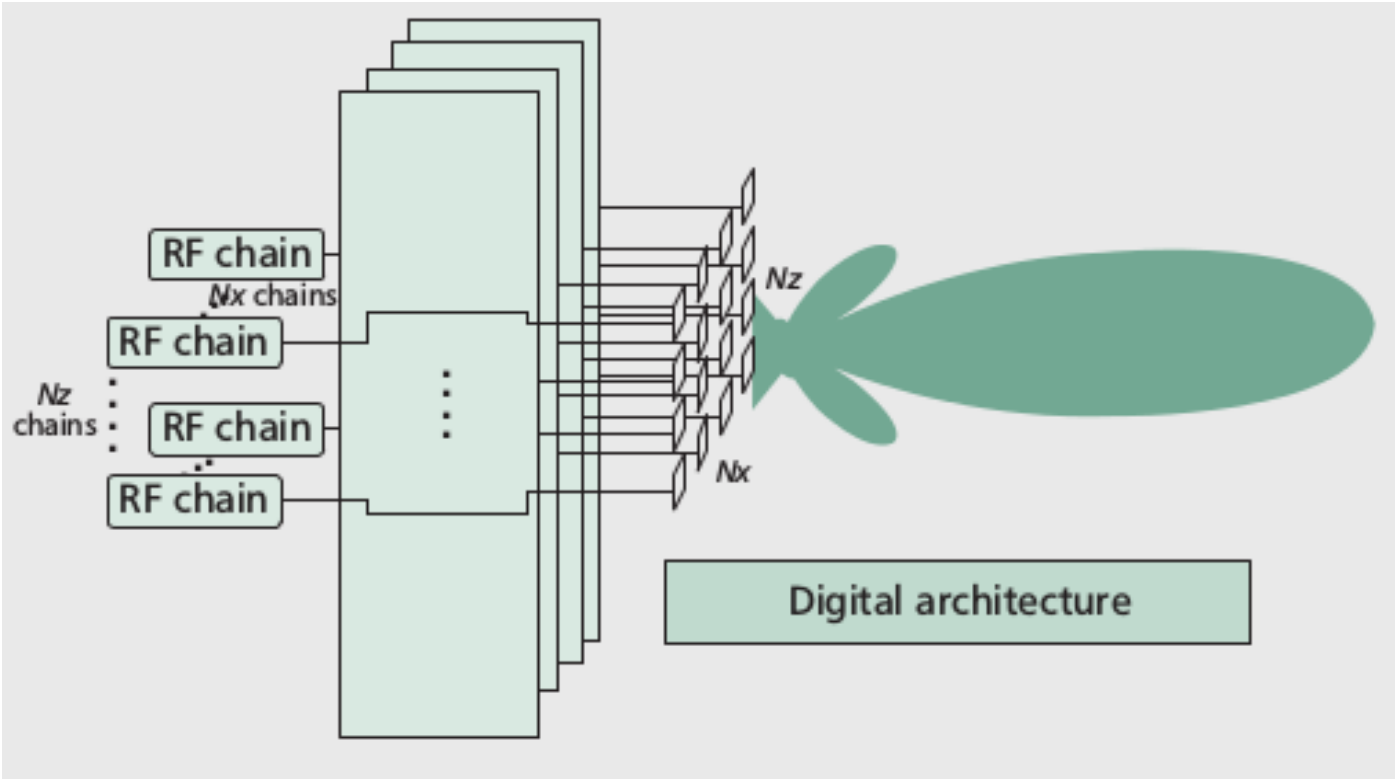
mmWave Architecture: Analog Beamforming



- Constant gain, and phased shifters are used
- Power consumption depends on the resolution of the quantized angles

Image source: http://www.profheath.org/research/millimeter-wave-cellular-systems

Digital Beamforming



In this design, both amplitude and phase are controlled digitally

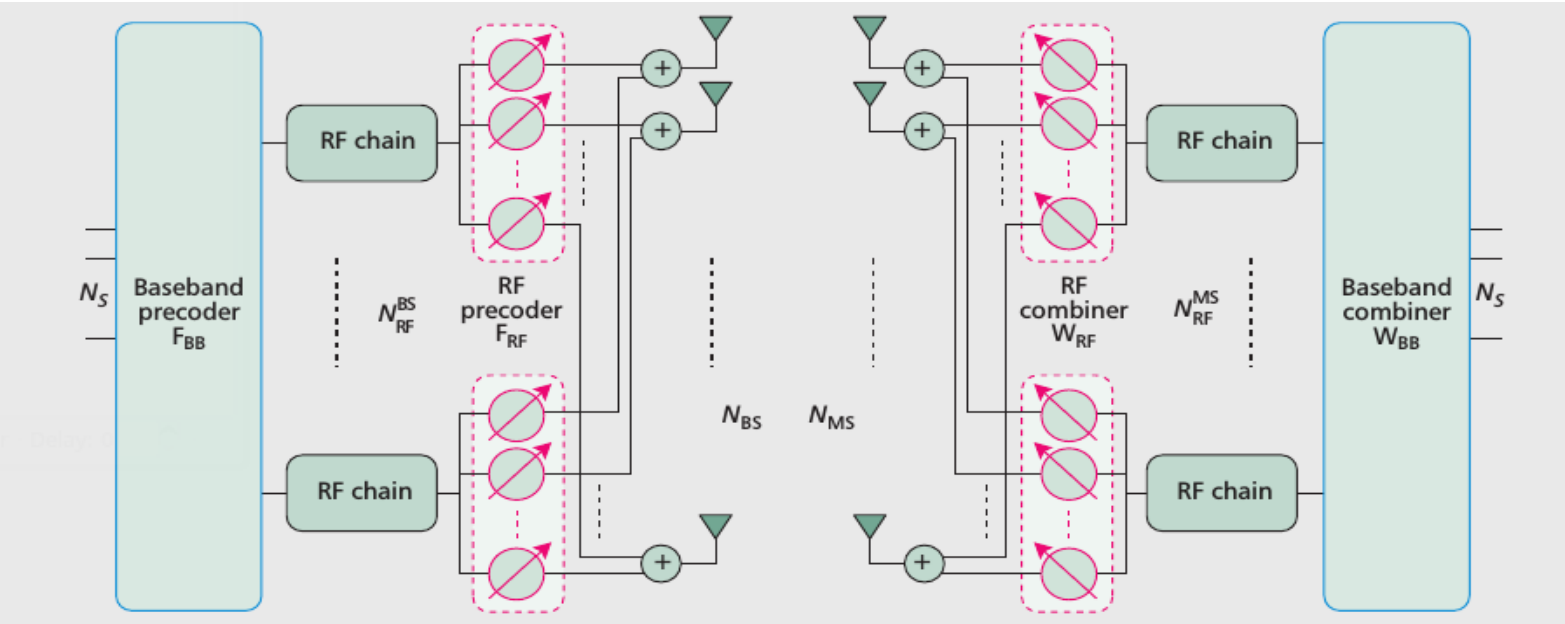
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Image source: Shu Sun, et al. "MIMO for Millimeter-Wave Wireless Communications: Beamforming, Spatial Multiplexing, or Both?", Comm. Magazine.

mmWave Architecture

- Conventional MIMO leverages heavily on digital signal processing
  - Dedicated number of RF chains (ADCs) for every antenna element
- Large number of antennas are employed at mmWave frequencies
  - Employing RF chains per antenna would incur more cost and complexity
- Analog signal processing along with digital (hybrid signal processing) poses a tenable solution at mmWave

Hybrid Beamforming



- Hybrid beamforming reduces the number of RF chains
- Digital can correct the analog limitations

Image source: Ahmed Alkhateeb, et al. "MIMO Precoding and Combining Solutions for Millimeter-Wave Systems", IEEE Comm. Magazine.

Design and Challenges

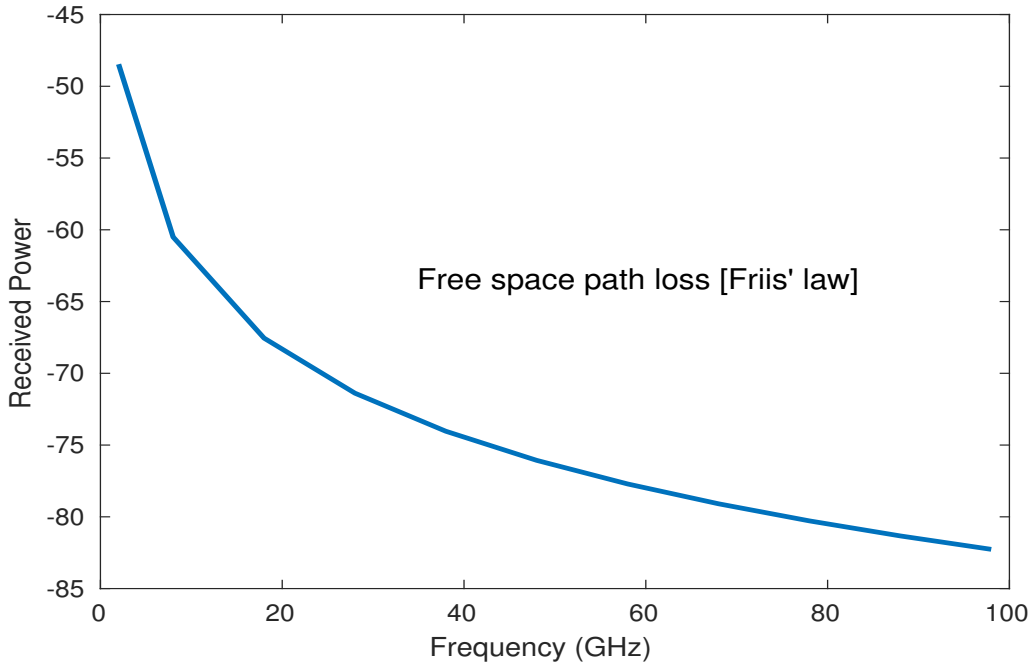
- Low-Complexity Precoding/Combining Designs
  - Difficulty comes from the coupling b/w analog and digital precoders
  - Constant modulus constraint on the phase shifters
- Channel Estimation with Hybrid Precoding
  - Channel matrix is large due to use of large arrays
  - Channel seen in the baseband is through the lens of the RF precoding and combining which complicates the training signal design
- Combining with low -resolution ADCs
  - ADCs with low resolution can achieve much of the unquantized capacity in the low and medium SNR regimes[1]

[1] J. Singh, O. Dabeer, and U. Madhow, On the Limits of Communication with Low-Precision Analog-to-Digital Conversion at the Receiver, IEEE Trans. Commun., vol. 57, no. 12, 2009, pp. 362939.

Study on Large Scale Parameters

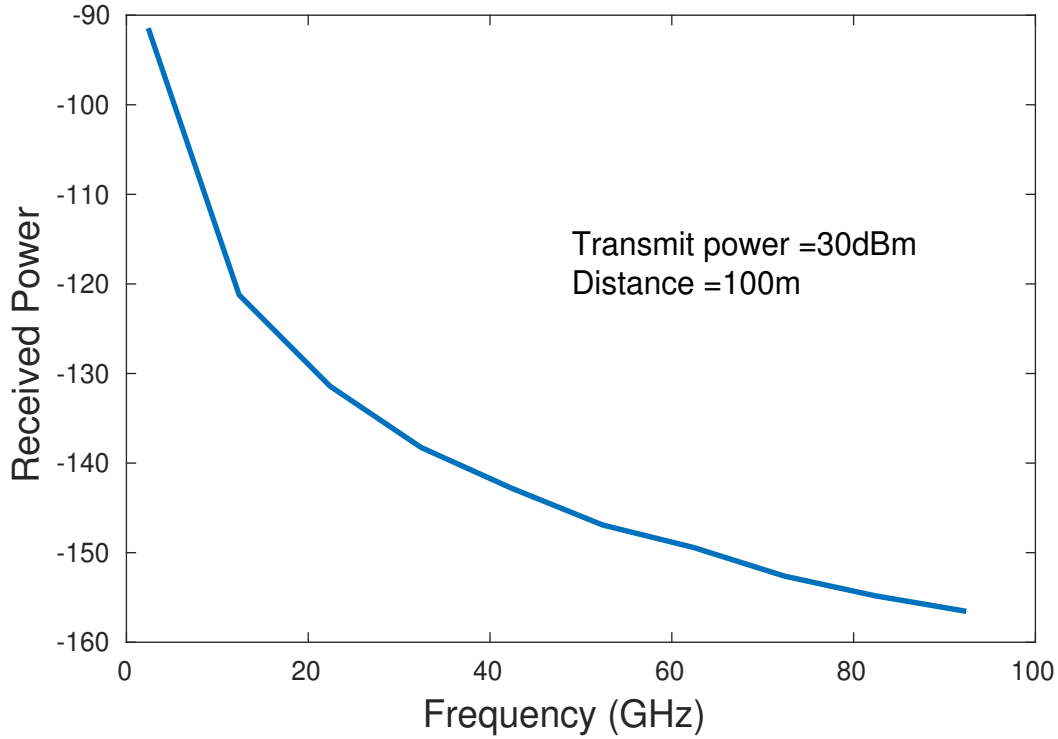
Trasmit power =30dBm

Distance =100m



Friis' law

$$P_r = G_r G_t \left(\frac{\lambda}{4\pi d}\right)^2 P_t$$



Single frequency close-in (CI) Model:

$$\mathrm{PL}^{\mathrm{CI}}(f,d)[\mathrm{dB}] = \mathrm{FSPL}(f,1\ \mathrm{m})[\mathrm{dB}] + 10n\log_{10}(d) + \chi_{\sigma}^{\mathrm{CI}},\ \text{where } d \geq\ 1\ \mathrm{m}$$

n is the path loss exponent

$n \approx 2.6$ LOS $\forall f$ in urban environment

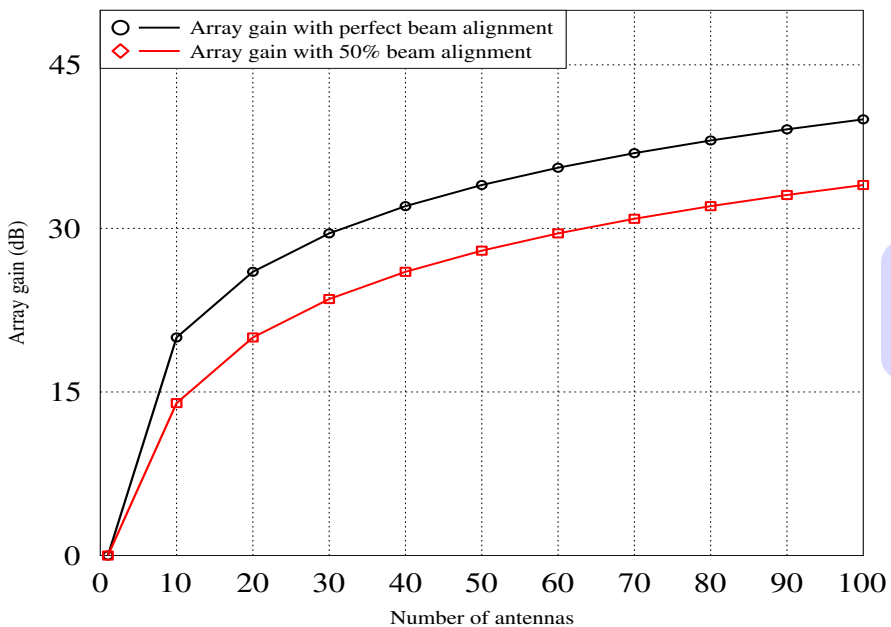
Antenna Design

Array Factor

$$\mathrm{AF} = 1 + e^{j\psi} + \ldots + e^{j(N-1)\psi}$$

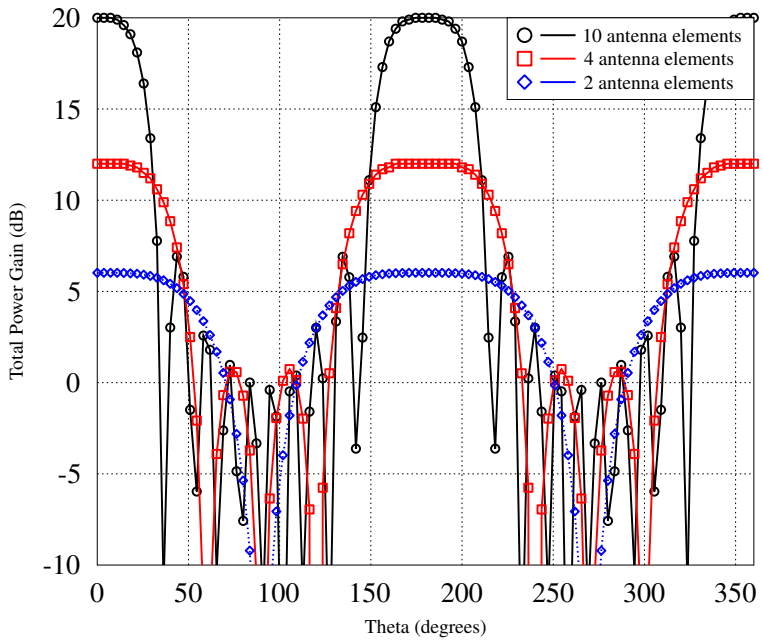
Where

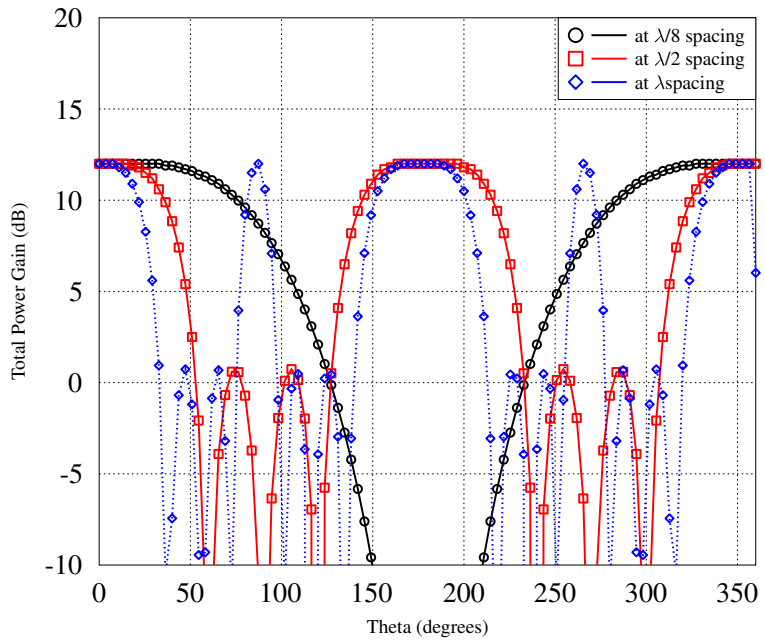
$$\psi = (2\pi/\lambda)\mathrm{d}cos(\theta) + \beta$$



$$G_{\mathrm{total}} = G_{\mathrm{element}} \mid \mathrm{AF} \mid^2$$

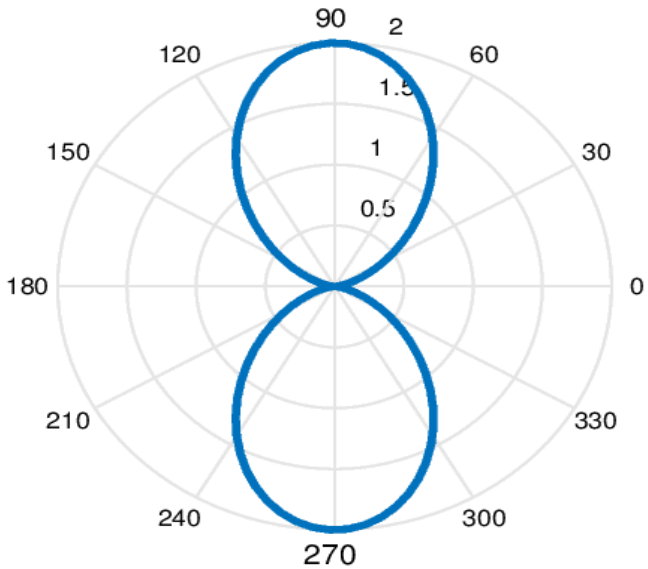
Radiation Pattern

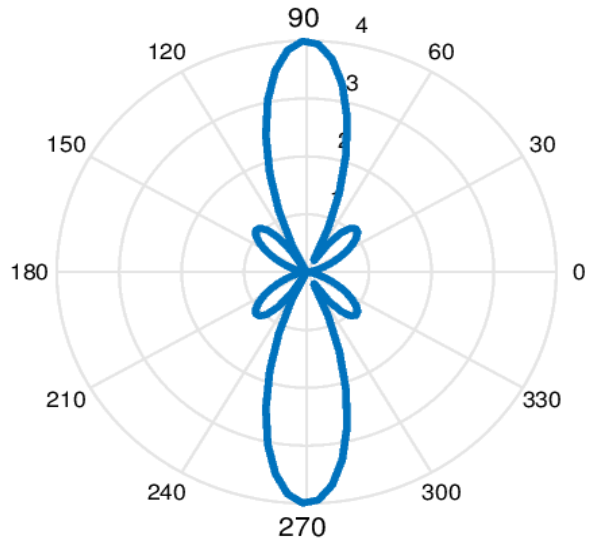




1. With increase in antenna elements, beamwidth decreases as observed in the first plot
2. In second figure, grating lobes are observed when spacing b/w antennas is $> \lambda/2$

Contd...

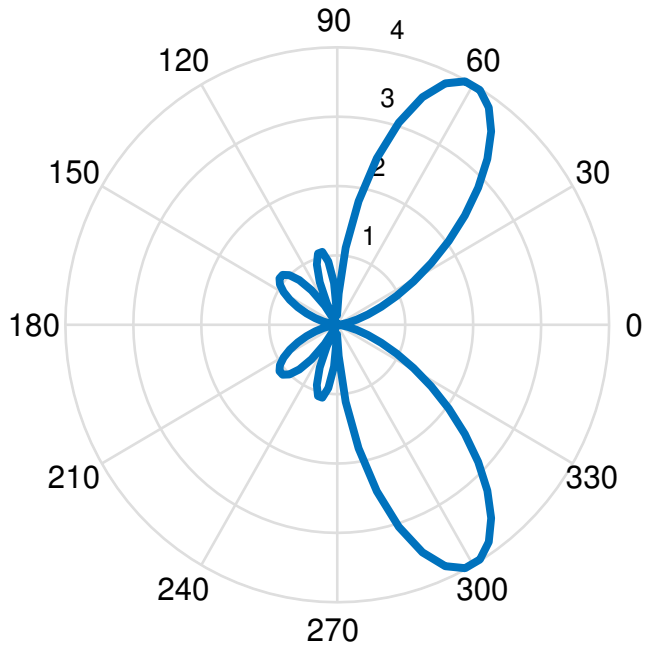




Beamwidth at $\lambda/2$ spacing with 2 and 4 antennas

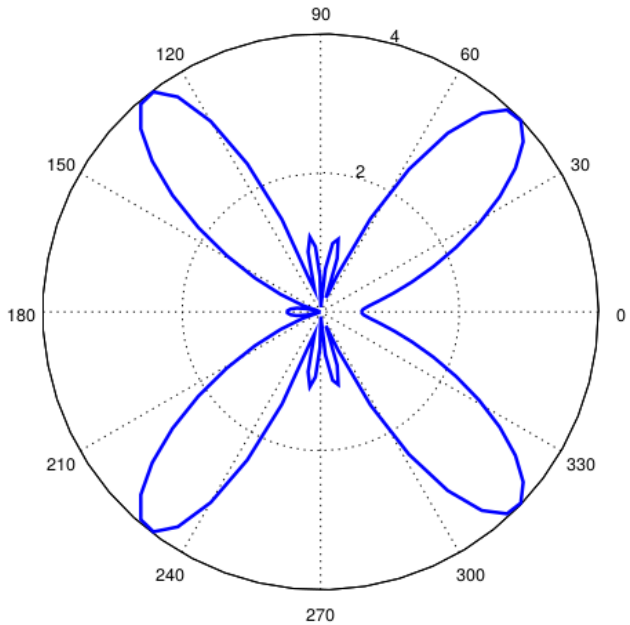
With increase in antenna elements beamwidth becomes narrower

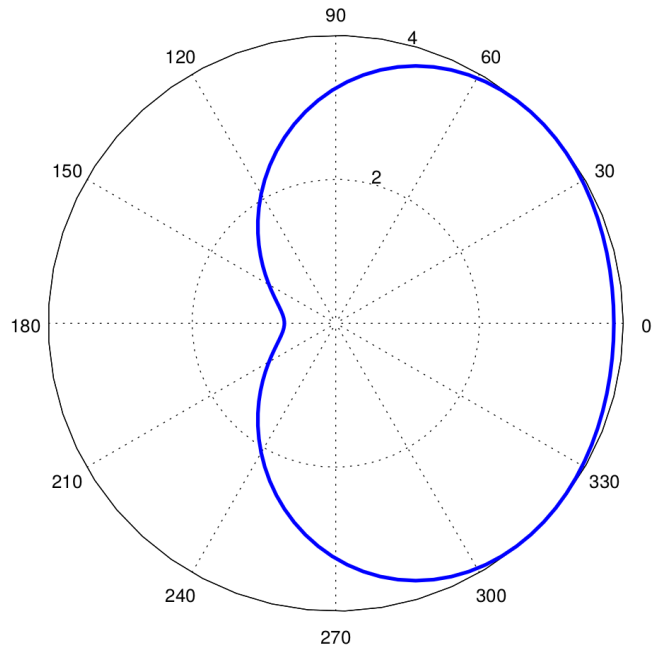
1. Beam steering along a direction



Beampattern along 60 degrees

Spacing Between Antennas





Beampattern at $3\lambda/4$ and $\lambda/8$ spacing between the antennas

- For $d > \lambda/2$, grating lobes are observed
- For $d < \lambda/2$, wider beamwidth is observed

Atmospheric Attenuation

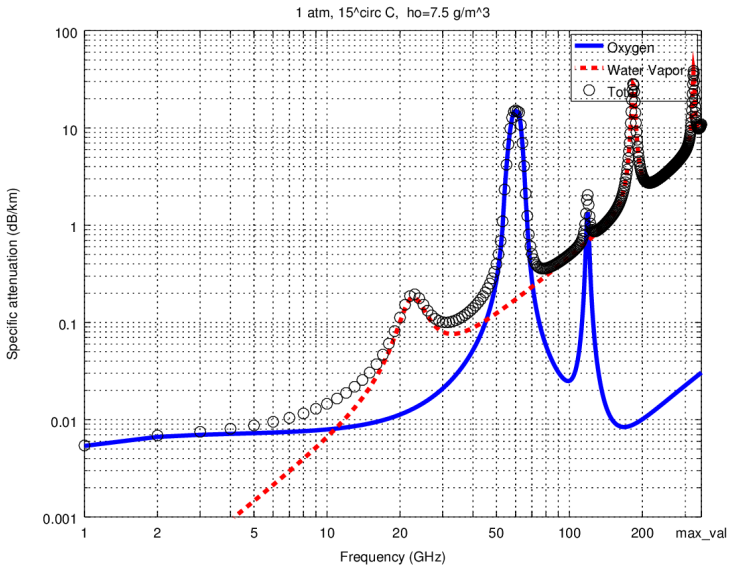


Figure: Atmospheric attenuation O_2 and H_2O

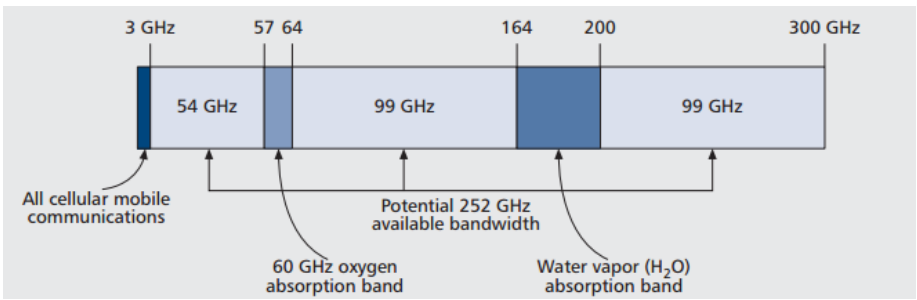
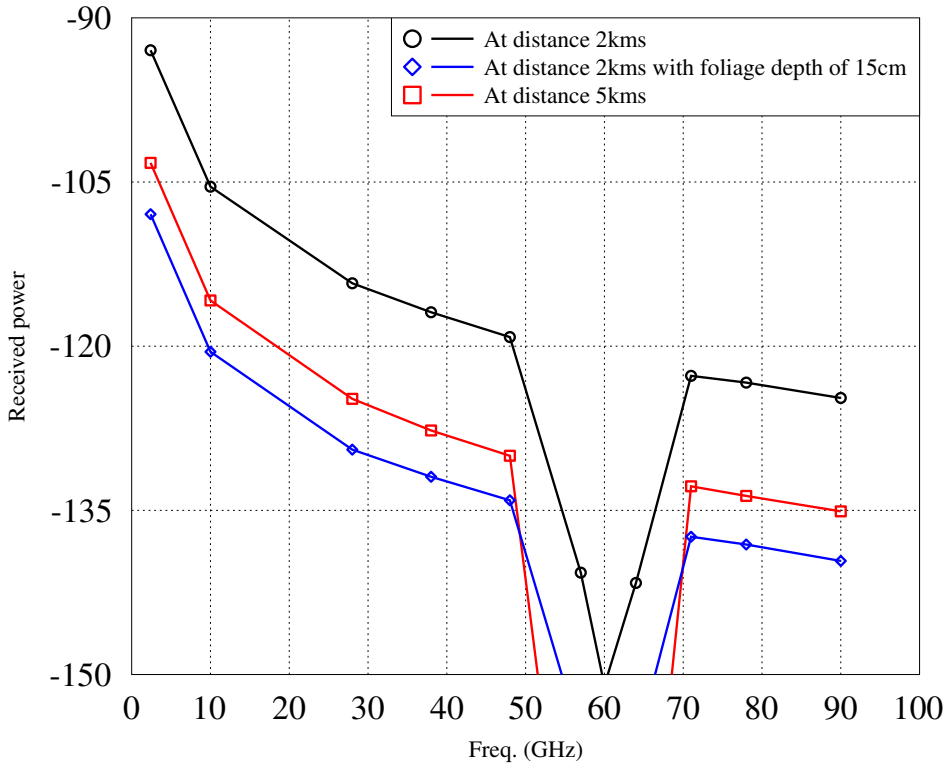


Figure: (b) mmWave spectrum

Source: Fig. (b) Zhouyue Pi and Farooq Khan, An Introduction to Millimeter-Wave Mobile Broadband Systems

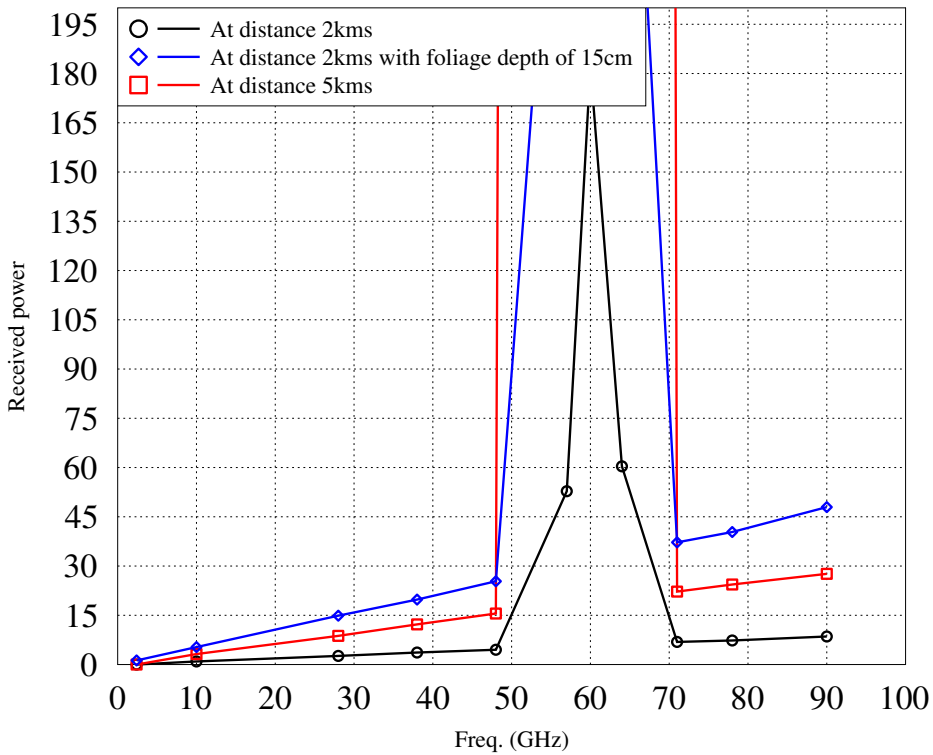
Path Loss



- Around 15-20 dB loss due to foliage (at foliage depth of 15cm)
- More suitable in urban environment
- Freq. < 50 GHz and > 70 GHz pose tenable solution for high data rates

Figure: Path loss with shadowing and atmospheric absorption

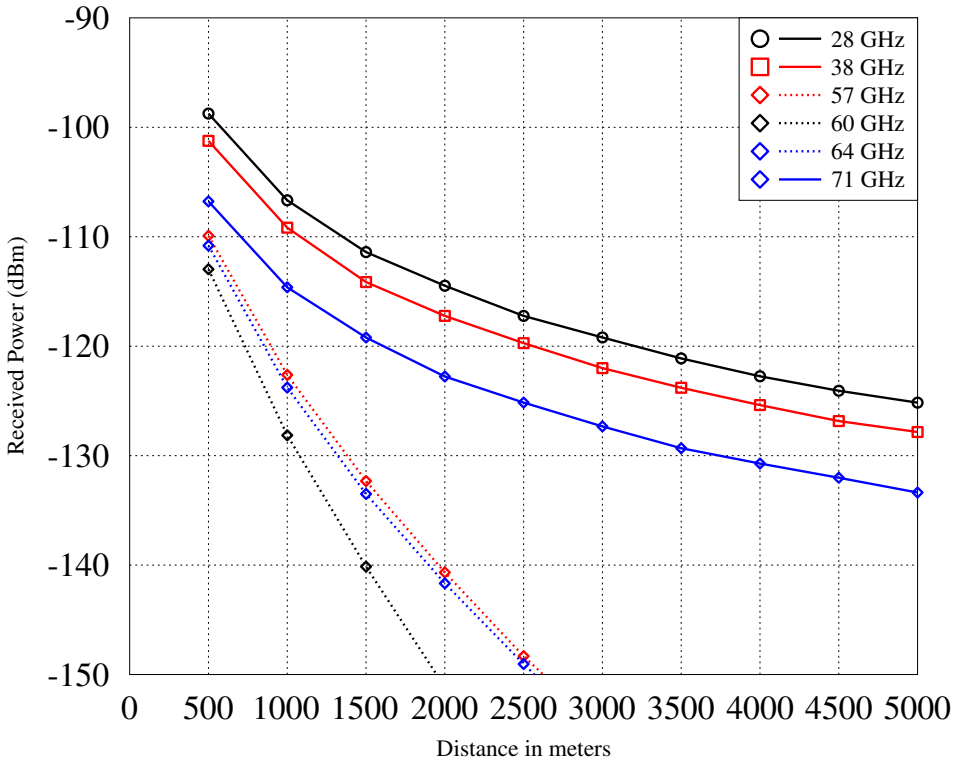
How Many Antennas?



- Min. 16 antennas are required at 38 GHz to achieve the required min. power (-104dBm) at 5km distance
- No. of antennas required increases almost linearly with frequency

Figure: No. of antenna elements required to achieve -104dBm

Path Loss with Distance



- Around 20dB more power is required to achieve -104dBm at 5kms for 28 and 38 GHz
- Using antenna arrays can compensate this path loss

Figure: No. of antenna elements required to achieve -104dBm

How Many Antennas with Distance?

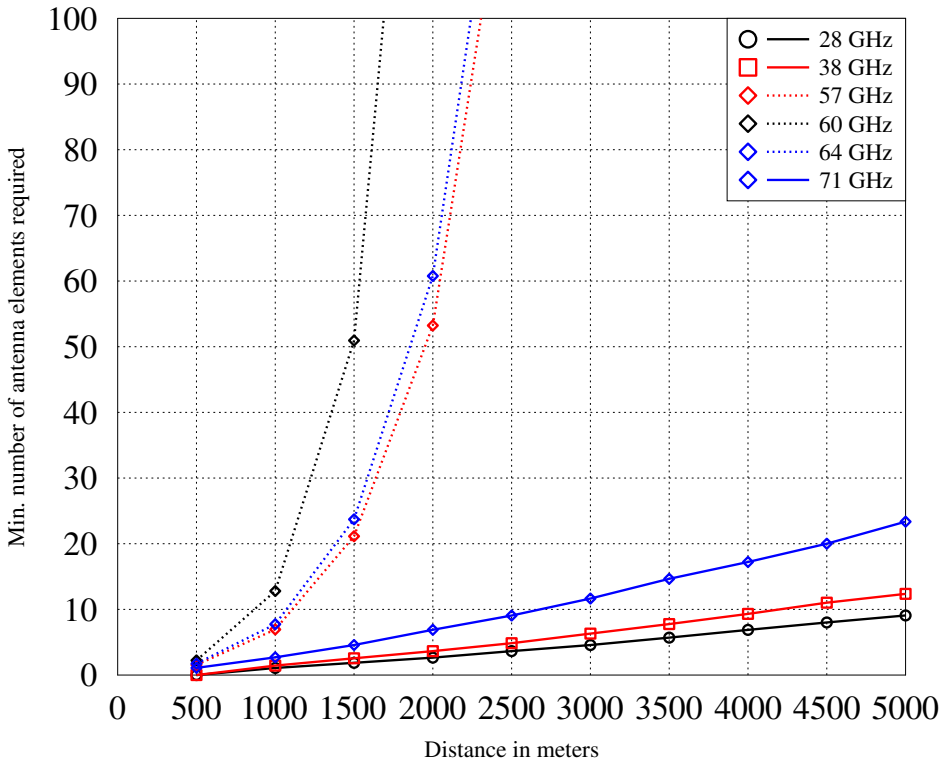


Figure: Received power vs distance

- Employing little more than few antennas can increase the coverage distance
- Around 10 and 25 min. antennas are required at 28 and 71 GHz respectively, for as large as 5km coverage

Doppler Spread

- The Doppler of a wireless channel depends on the carrier frequency and mobility
- In rich scattering and omnidirectional scenario, the max. Doppler shift for carrier frequency of 3-60 GHz with mobility of 3-350 km/h ranges from 10 Hz to 20 kHz
- The Doppler shift values of incoming waves on different angles at the receiver are different, resulting in a phenomenon called Doppler spread
- In mmWave, the narrow beams will significantly reduce angular spread of the incoming waves, which in turn reduces the Doppler spread

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Zhouyue Pi and Farooq Khan, An Introduction to Millimeter-Wave Mobile Broadband Systems, IEEE Communications Magazine, June 2011

Hybrid Precoding

Spatially Sparse Precoding in Millimeter Wave MIMO Systems

Omar El Ayach, Sridhar Rajagopal, Robert W. Heath Jr, et al.

TWC'14

Goal: To formulate the precoding/combining problem as a sparse reconstruction problem

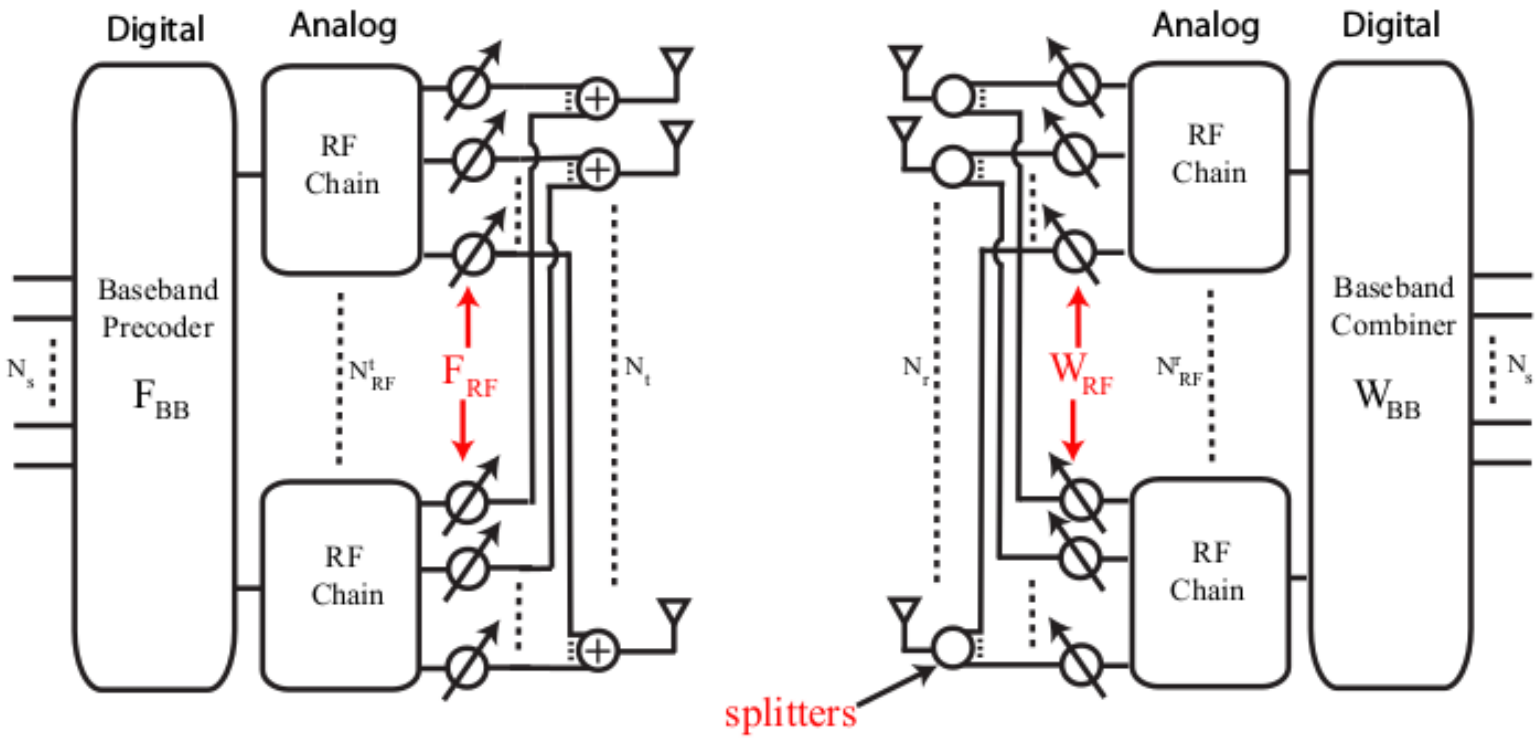
Model: Single user with N_t and N_r antennas such that $N_s \leqslant N_t^{RF} \leqslant N_t$

$$Y = \sqrt{\rho}\mathbf{H}\mathbf{F}_{\mathrm{RF}}\mathbf{F}_{\mathrm{BB}}s + n$$

such that, $\|F_{\mathrm{RF}}F_{\mathrm{BB}}\|^2 = N_s$, and $F_{\mathrm{RF}} \in \mathcal{F}_{\mathrm{RF}}$

Contributions:

- Using the principle of basis pursuit, authors develop algorithms that accurately approximate optimal unconstrained precoders and combiners such that they can be implemented in low-cost RF hardware.



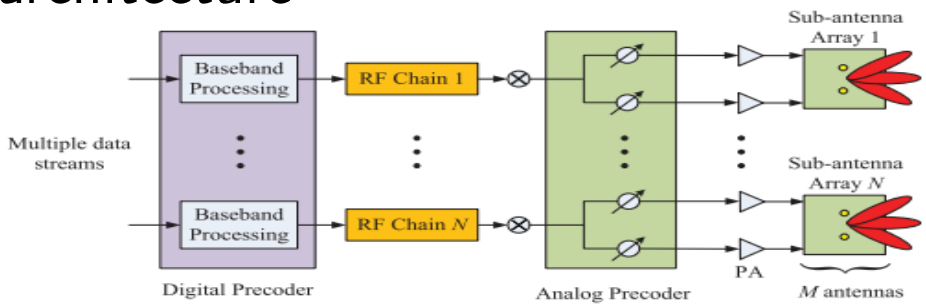
- To design combiners that minimize the MSE b/w the transmitted and processed received signals
- Showed that the proposed precoders can be efficiently quantized and that the precoding strategy is well-suited for limited feedback systems

Energy-Efficient Hybrid Analog and Digital Precoding for MmWave MIMO Systems With Large Antenna Arrays

Xinyu Gao, Robert W. Heath

JSAC'16

Goal: To design more energy efficient hybrid precoding with subconnected architecture



Contributions & Conclusion:

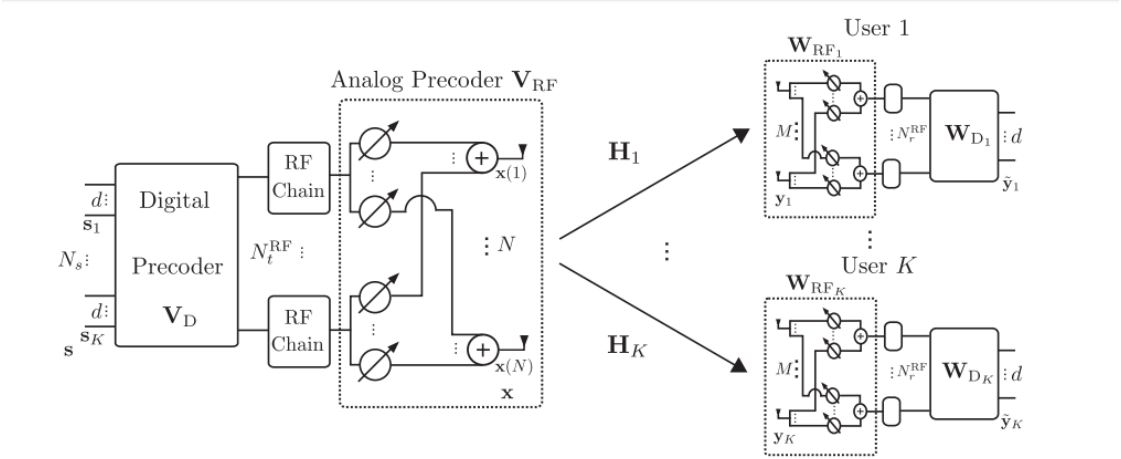
- SIC-based precoding, which avoids the SVD and matrix inversion
- Only about 10% as complex as that of previously designed precoding
- Higher energy efficiency (bps/Hz/W) than both the spatial sparse precoding and the fully digital precoding

Hybrid Digital and Analog Beamforming Design for Large-Scale Antenna Arrays

Foad Sohrabi, Wei Yu

JSTSP'16

Goal: To propose an heuristic algorithms that can approach the performance of a fully digital scheme with much fewer number of RF chains.



Proposition: With $N_{\mathrm{RF}} \geq 2N_s$ any fully digital beamforming can be realized

Contributions: Proposed hybrid beamforming algorithm for solving the overall spectral efficiency maximization

Subspace Estimation and Decomposition for Large Millimeter-Wave MIMO Systems

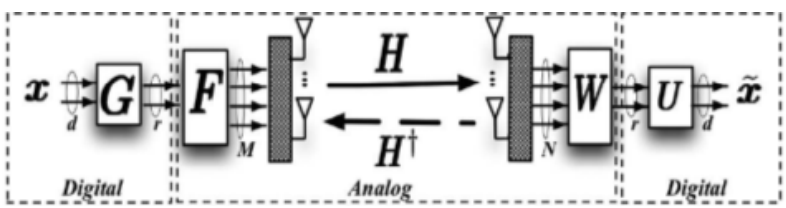
Hadi Ghauch, et al.

JSTSP'16

Goal: To estimate the right (resp. left) singular subspaces of the channel, at the BS (resp. MS)

Contributions:

- Proposed a method (based on the well-known Arnoldi iteration) exploiting channel reciprocity in TDD systems and the sparsity of the channels eigenmodes
- Bounds on estimation error in presence of distortion at BS (MS) is proposed
- Also, proposed an iterative algorithm for subspace decomposition by a cascade of hybrid analog-digital architecture



Hybrid Precoding Design in Millimeter Wave MIMO Systems: An Alternating Minimization Approach

Xianghao Yu (Mediatek), K. B. Letaief, et al.

JSTSP'16

Goal: To design the hybrid precoders using alternating minimization approach in mmWave MIMO systems

Contributions:

- Identification of manifold (Riemannian manifold) structure in anlog precoding
- Powerful manifold optimization tool is applied

Conclusions:

- Provides higher spectral efficieny than the hitherto existing methods
- Fully connected architecture performs better than partially connected system
- Computation efficiency will need to be improved to make it practical

Iterative Eigenvalue Decomposition and Multipath-Grouping Tx/Rx Joint Beamformings for Millimeter-Wave Communications

Zhenyu Xiao, Xiang-Gen Xia, et al.

TWC'15

Goal: To design Tx/Rx joint beamforming schemes in MMWC

Contributions:

- Two Tx/Rx joint beamforming schemes have been proposed
  1. The first one is IEVD, which is suboptimal and suitable under slow fading channel.
  2. The other one is MPG suitable under fast fading channel, which groups the MPCs and concurrently beamforms towards multiple steering angles of the grouped MPCs

Conclusions:

- MPG achives full diversity, and better array gain
- Array weights are obtained for every MPC
- EVD involved high complexity in channel training

Other papers

1. Weiheng Ni, "Near-Optimal Hybrid Processing for Massive MIMO Systems via Matrix Decomposition", arXiv:1504.03777v1

Notes